

|                                                         | $\mathcal{K}_{0^+}^{\#1}$                         | $\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$     |                                           |                                               |                                      |                                      |
|---------------------------------------------------------|---------------------------------------------------|---------------------------------------------------|-------------------------------------------|-----------------------------------------------|--------------------------------------|--------------------------------------|
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger$                     | 0                                                 | 0                                                 |                                           |                                               |                                      |                                      |
| $\mathcal{K}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                                 | $Y_1 k^2 + 3 \overset{(2)}{K}_1$                  | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$     | $\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$ | $\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$ |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $Y_2 k^2 + 2 \overset{(2)}{K}_1$                  | $\frac{Y_2 k^2 + 2 \overset{(2)}{K}_1}{\sqrt{2}}$ | 0                                         | 0                                             |                                      |                                      |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $\frac{Y_2 k^2 + 2 \overset{(2)}{K}_1}{\sqrt{2}}$ | $\frac{1}{2} (Y_2 k^2 + 2 \overset{(2)}{K}_1)$    | 0                                         | 0                                             |                                      |                                      |
| $\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                                 | 0                                                 | $\frac{Y_3 k^2}{2}$                       | $-\frac{Y_4 k^2}{2 \sqrt{2}}$                 |                                      |                                      |
| $\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                                 | 0                                                 | $-\frac{Y_4 k^2}{2 \sqrt{2}}$             | $\frac{Y_5 k^2}{2}$                           |                                      |                                      |
|                                                         |                                                   |                                                   | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |                                      |                                      |
| $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     |                                                   |                                                   | 0                                         | 0                                             |                                      |                                      |
| $\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ |                                                   |                                                   | 0                                         | $\frac{Y_6 k^2}{2}$                           |                                      |                                      |

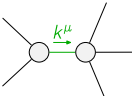
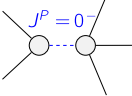
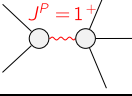
|                                                         | $\mathcal{J}_{0^+}^{\#1}$            | $\mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                              |                                               |                                      |                                      |
|---------------------------------------------------------|--------------------------------------|-----------------------------------------------|----------------------------------------------|-----------------------------------------------|--------------------------------------|--------------------------------------|
| $\mathcal{J}_{0^+}^{\#1}{}^\dagger$                     | 0                                    | 0                                             |                                              |                                               |                                      |                                      |
| $\mathcal{J}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                    | $\frac{1}{\text{Det}(0^+)}$                   | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$    | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$     | $\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$ | $\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$ |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{2}{3 \text{Det}(1^+)}$        | $\frac{\sqrt{2}}{3 \text{Det}(1^+)}$          | 0                                            | 0                                             |                                      |                                      |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $\frac{\sqrt{2}}{3 \text{Det}(1^+)}$ | $\frac{1}{3 \text{Det}(1^+)}$                 | 0                                            | 0                                             |                                      |                                      |
| $\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$          | 0                                    | 0                                             | $\frac{Y_5 k^2}{2 \text{Det}(1^+)}$          | $\frac{Y_4 k^2}{2 \sqrt{2} \text{Det}(1^+)}$  |                                      |                                      |
| $\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$          | 0                                    | 0                                             | $\frac{Y_4 k^2}{2 \sqrt{2} \text{Det}(1^+)}$ | $\frac{Y_3 k^2}{2 \text{Det}(1^+)}$           |                                      |                                      |
|                                                         |                                      |                                               | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$    | $\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |                                      |                                      |
| $\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     |                                      |                                               | 0                                            | 0                                             |                                      |                                      |
| $\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ |                                      |                                               | 0                                            | $\frac{1}{\text{Det}(2^+)}$                   |                                      |                                      |

#### Abbreviations used in matrices

$$\begin{aligned}
Y_1 &= \overset{(4)}{K}_{15} - \overset{(4)}{K}_{16} \&\& Y_2 = 2 \overset{(4)}{K}_{15} - \overset{(4)}{K}_3 \&\& Y_3 = 2 (\overset{(4)}{K}_1 + \overset{(4)}{K}_{15}) + \overset{(4)}{K}_{16} \&\& Y_4 = 2 \overset{(4)}{K}_1 - \overset{(4)}{K}_7 \&\& \\
Y_5 &= \overset{(4)}{K}_1 - 2 \overset{(4)}{K}_{15} - \overset{(4)}{K}_{16} - \overset{(4)}{K}_7 \&\& Y_6 = 2 \overset{(4)}{K}_{15} + \overset{(4)}{K}_{16} \&\& \text{Det}(0^+) = k^2 (\overset{(4)}{K}_{15} - \overset{(4)}{K}_{16}) + 3 \overset{(2)}{K}_1 \&\& \\
\text{Det}(1^+) &= k^2 (3 \overset{(4)}{K}_{15} - \frac{3 \overset{(4)}{K}_3}{2}) + 3 \overset{(2)}{K}_1 \&\& \text{Det}(2^+) = \frac{1}{2} k^2 (2 \overset{(4)}{K}_{15} + \overset{(4)}{K}_{16})
\end{aligned}$$

#### Lagrangian

$$\begin{aligned}
&\overset{(2)}{K}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{K}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{K}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} - \overset{(4)}{K}_1 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} + \\
&\overset{(4)}{K}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + 4 \overset{(4)}{K}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + 2 \overset{(4)}{K}_{16} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \\
&2 \overset{(4)}{K}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + 2 \overset{(4)}{K}_{15} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \overset{(4)}{K}_{16} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} - \overset{(4)}{K}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} + \\
&\overset{(4)}{K}_7 \partial^\chi \mathcal{K}^\alpha_{\alpha\beta} \partial_\delta \mathcal{K}^\delta_{\chi\beta} - \frac{1}{2} \overset{(4)}{K}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \overset{(4)}{K}_{15} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{K}_{16} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

|                                                                                    |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_{0^+}^{\#1} = 0$                                                      | 1                                                             | $\partial_\beta \mathcal{J}^\alpha_{\alpha\beta} = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_{1^+}^{\#1\alpha\beta} = \mathcal{J}_{1^+}^{\#2\alpha\beta}$          | 3                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} = 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta} = 0$                                           | 5                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\chi \mathcal{J}^\chi_{\chi}{}^\delta = 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi_{\chi}{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Total # constraint(s):                                                             | 9                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Residue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $-\frac{4 \overset{(4)}{K}_{15}{}^2 + \overset{(4)}{K}_{16}{}^2 + \overset{(4)}{K}_{16} \overset{(4)}{K}_7 + \overset{(4)}{K}_7{}^2 + 2 \overset{(4)}{K}_{15} (2 \overset{(4)}{K}_{16} + \overset{(4)}{K}_7)}{(2 \overset{(4)}{K}_{15} + \overset{(4)}{K}_{16}) (8 \overset{(4)}{K}_{15}{}^2 + 2 \overset{(4)}{K}_{16}{}^2 + 2 \overset{(4)}{K}_1 (2 \overset{(4)}{K}_{15} + \overset{(4)}{K}_{16}) + 2 \overset{(4)}{K}_{16} \overset{(4)}{K}_7 + \overset{(4)}{K}_7{}^2 + 4 \overset{(4)}{K}_{15} (2 \overset{(4)}{K}_{16} + \overset{(4)}{K}_7))}$ |
|  | 1                                                             | $\frac{3 \overset{(2)}{K}_1}{-\overset{(4)}{K}_{15} + \overset{(4)}{K}_{16}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\frac{1}{-\overset{(4)}{K}_{15} + \overset{(4)}{K}_{16}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | 3                                                             | $\frac{2 \overset{(2)}{K}_1}{-2 \overset{(4)}{K}_{15} + \overset{(4)}{K}_3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\frac{2}{6 \overset{(4)}{K}_{15} - 3 \overset{(4)}{K}_3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Resolved unitarity condition(s):                                                   | (Demonstrably impossible)                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |